

Winding Tree

Market Protocol SDK BETA is live!



Mokhi · Following

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We are thrilled to share with you an exciting development that promises to revolutionize the travel industry as we know it. This month marked a momentous occasion with the first public release of the protocol SDK. For those unfamiliar with the jargon, SDK stands for Software Development Kit, a comprehensive set of tools and code that developers can use to build applications on top of a particular platform. And in this case, the platform is the Winding Tree blockchain, a decentralized travel marketplace.

So, what does this mean for travelers and blockchain enthusiasts alike? Let's break it down in simple "human" terms:

1. The Winding Tree Protocol: Empowering the Future of Travel

The Winding Tree protocol is a decentralized ecosystem that connects travelers directly with suppliers, cutting out intermediaries and reducing costs. Imagine a travel marketplace where hotels, airlines, car rental companies, and other service providers can interact directly with travelers, offering competitive prices and personalized experiences. No more hidden

fees or long booking processes — just a seamless, transparent, and efficient travel experience.

2. The SDK Release: Making Blockchain Accessible

With the recent SDK release, we made significant strides in stabilizing our codebase and making it available to developers and businesses. The SDK is a collection of powerful tools and utilities that simplify the development process and allow seamless integration of the Winding Tree protocol into applications.

3. What's in the SDK BETA?

The SDK comprises several packages, each serving a specific purpose:

- **@windingtree/contracts:** This package contains the smart contracts and utilities that underpin the Winding Tree market protocol.
- **@windingtree/sdk-server:** A coordination server built on top of the libp2p library, enabling fast and secure peer-to-peer messaging for efficient communication between buyers and suppliers.
- **@windingtree/sdk-node:** An enhanced version of the protocol client, designed to establish connections with the coordination server while catering to specific supplier use cases. This node comes equipped with additional features and functionalities, empowering it to actively and effectively participate within the protocol ecosystem. By leveraging these enhancements, suppliers can optimize their interactions and operations within the Winding Tree market protocol network.
- **@windingtree/sdk-node-api:** A powerful toolset that enables remote management of the protocol node. Built on the foundation of tRPC, an end-to-end typesafe RPC framework, this API facilitates seamless user management, admin management, and deal management procedures. With its comprehensive functionality, users can efficiently handle various tasks related to node administration and coordination, providing

a robust and user-friendly experience for managing the protocol node remotely.

- **@windingtree/sdk-client:** A client of the coordination server with essential tools for creating, validating, and publishing requests, and catching incoming offers.
- **@windingtree/sdk-react:** This package provides utilities for easy integration of protocol features into React-based applications, empowering developers to enhance their projects with seamless protocol integration.
- **@windingtree/sdk-constants:** Useful constants that can be used for the protocol components configuration
- **@windingtree/sdk-types:** A collection of TypeScript interfaces and types which are commonly used across the Winding Tree SDK. It includes data types for messages, requests, offers, deals, contracts, and various configuration options.
- **@windingtree/sdk-utils:** A collection of helper utilities that are used throughout the Winding Tree SDK. It includes functions for working with H3 spatial indexing, text encoding and decoding, time calculations and conversions, generating unique identifiers (UIDs), and Ethereum wallet operations.
- **@windingtree/sdk-test-utils:** A utility package designed for testing purposes within the Winding Tree SDK. It extends the capabilities of the vitest test runner with a custom function for validating deep equality of objects, more features will be added later.
- **@windingtree/sdk-pubsub:** A centralized publish/subscribe protocol (CenterSub) for peer-to-peer communication within a WindingTree market protocol network, built on top of the libp2p-gossipsub protocol.
- **@windingtree/sdk-contracts-manager:** A package that provides an interface to interact with the protocol smart contracts, allowing developers to execute various operations including transactions and contract function calls.

- **@windingtree/sdk-messages:** This package offers a way to build, sign and verify offers and requests for Winding Tree market protocol. It takes care of data validation, hashing, and signing/verification of EIP-712 signatures. Additionally, it provides utilities for test environments.
- **@windingtree/sdk-queue:** A cross-platform tasks queue system designed to manage jobs that need to be executed in an asynchronous manner.
- **@windingtree/sdk-storage:** A key-value database abstraction layer for Winding Tree SDK. It includes concrete implementations for in-memory storage and browser localStorage. More implementations are expected soon.
- **@windingtree/sdk-db:** A package that includes functionalities for managing users' and deals' data in a secure and effective way. Additionally, the package offers utilities for pagination.
- **@windingtree/sdk-logger:** A simple logging utility for Winding Tree SDK based on the debug package. The logger supports multiple logging levels and enables/disables logging for a particular named scope.

The SDK examples

Coordination Server:

<https://github.com/windingtree/sdk/tree/beta/examples/server>

The protocol coordination server example application (Node.js-based).

Protocol Client:

<https://github.com/windingtree/sdk/tree/beta/examples/client>

The protocol client is a React-based web application that allows users to send simple requests and obtain offers from suppliers in response and then make deals based on the offers.

Protocol Node:

<https://github.com/windingtree/sdk/tree/beta/examples/node>

A Node.js-based application that implements a basic set of the protocol Node features.

Node Manager:

<https://github.com/windingtree/sdk/tree/beta/examples/manager>

A React-based web application that allows users to register supplier entities in the protocol smart contract and provides simple supplier reception features, such as reviewing existing deals, and making user check-in and check-out operations.

What's Next for Winding Tree?

Looking ahead, Winding Tree has some exciting plans in the pipeline:

- **Improvement of e2e Test Suite:** This will ensure the platform's robustness and reliability, ensuring a smooth experience for both travelers and suppliers.
- **More Unit Tests:** Expanding unit testing will help identify and fix potential bugs early in the development process.
- **Blockchain Testing Network Integration:** This integration will further enhance the development environment and allow for the exploration of new test cases.
- **Enhancing @windingtree/sdk-react:** Winding Tree aims to empower developers with even more useful protocol client utilities to create better travel applications.

The release of the Winding Tree Protocol SDK is a significant milestone that brings us one step closer to a decentralized future for the travel industry. By connecting travelers directly with suppliers through blockchain technology, Winding Tree aims to create a fairer, more transparent, and more efficient travel marketplace.

Whether you're a travel enthusiast excited about the prospect of better travel experiences or a blockchain enthusiast eager to witness the transformative power of decentralized technologies, the Winding Tree protocol and its SDK have something to offer.

To learn more and access the protocol documentation, visit:
<https://windingtree.github.io/sdk>

So, let's embark on this journey together, as the future of travel unfolds before our eyes, driven by the innovation and possibilities of blockchain technology.

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